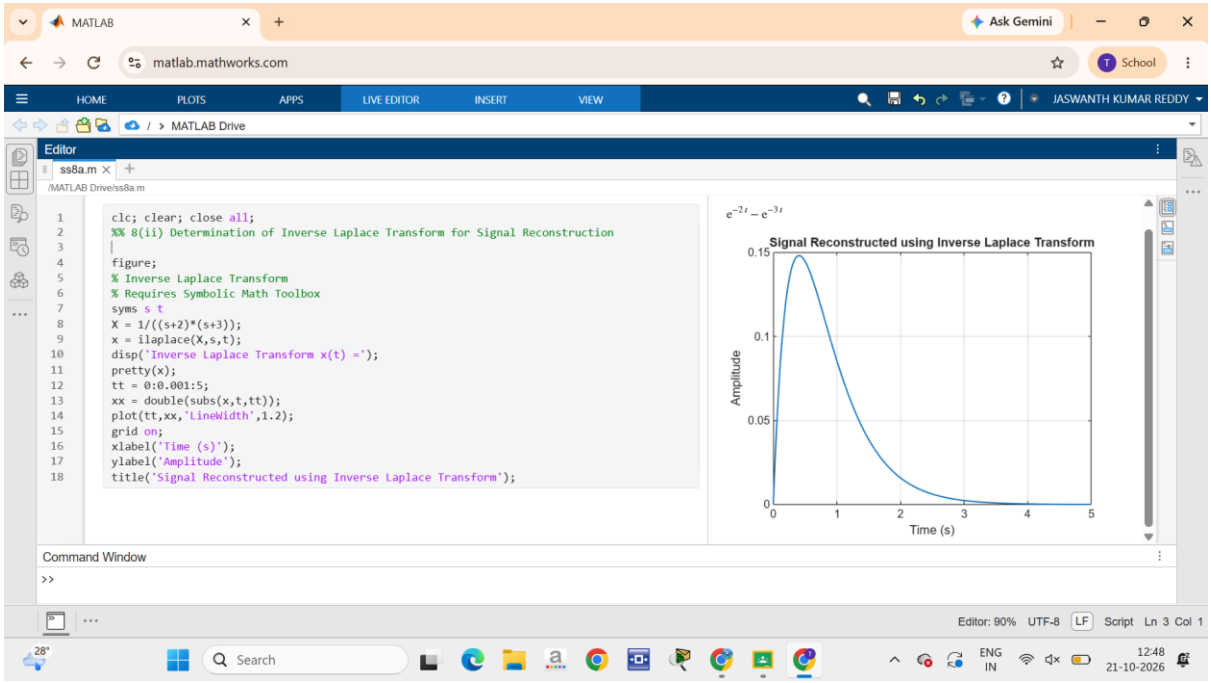
Experiment – 8:

Laplace Transform, Inverse Transform, ROC and Stability

OBJECTIVES:

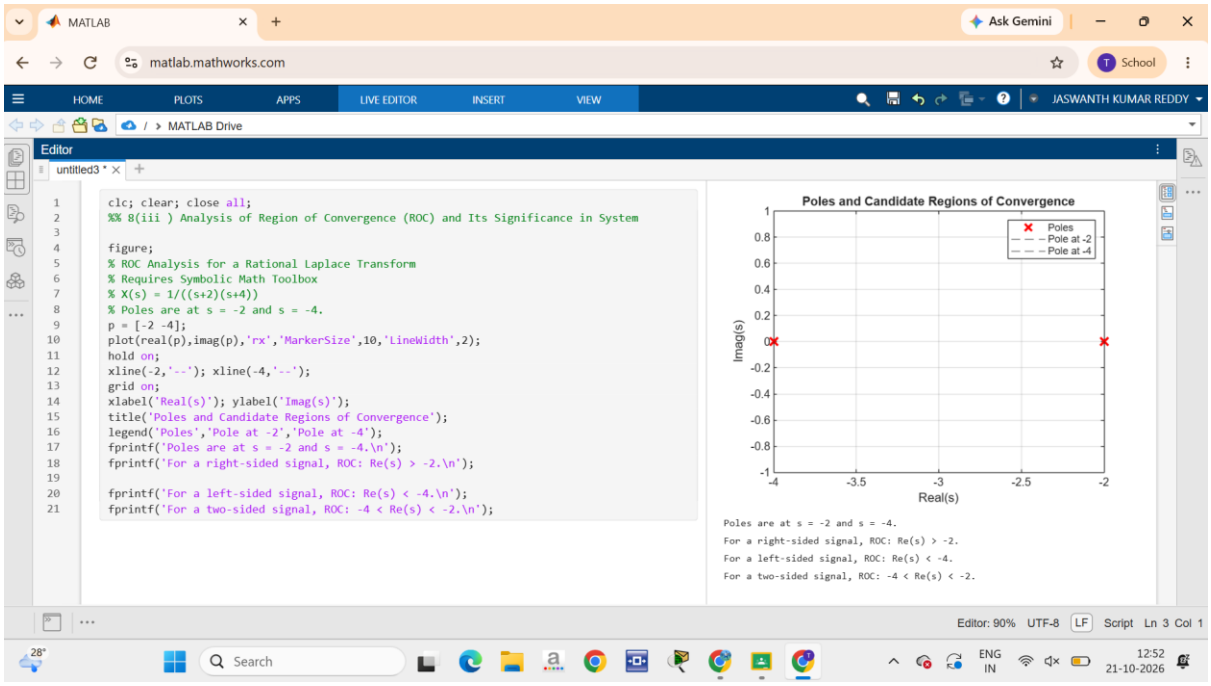
i) To determine the inverse Laplace transform of a given frequency-domain representation and reconstruct the corresponding time-domain signal.

Output Graphs:



ii) To determine the Region of Convergence (ROC) of Laplace transforms and analyze its significance in characterizing continuous-time signals and systems.

Output Graphs:



Result: Laplace transforms and inverse Laplace transforms were evaluated, and the region of convergence (ROC), poles, zeros, and system stability were analyzed.